

Homework 4 File Submission

Find the solution to each recurrence relation using the iterative approach. Show all work!

A. $a_n = 2a_{n-1} - 3, a_0 = -1$

$$= 2(2a_{n-2} - 3) - 3 = 2^2 a_{n-2} - 3(2+1)$$

$$= 2(2^2 a_{n-3} - 3(2+1)) - 3 = 2^3 a_{n-3} - 3(2^2 + 2 + 1)$$

$\vdots$

$$= 2^n a_{n-n} - 3(2^{n-1} + 2^{n-2} + \dots + 2 + 1)$$

$$= 2^n a_0 - 3(2^{n-1} + 2^{n-2} + \dots + 2 + 1)$$

$$= 2^n a_0 - 3(2^n - 1)$$

Sum of geometric series:

$$2^{n-1} + 2^{n-2} + \dots + 2 + 1 = \frac{2^n - 1}{2 - 1} = 2^n - 1$$

$$\begin{aligned} a_1 &= 2(-1) - 3 = -5 \\ a_2 &= 2(-5) - 3 = -13 \end{aligned}$$

Now plug in $a_0 = -1$:

$$a_n = 2^n(-1) - 3(2^n - 1)$$

$$= -2^n - 3(2^n) + 3$$

$$= -2^n(1+3) + 3$$

$$= 3 - 4(2^n)$$

$$= 3 - 2^2(2^n)$$

$$\boxed{= 3 - 2^{n+2}} \leftarrow \text{Final closed form}$$

$$\begin{aligned} a_0 &= 3 - 2^{0+2} = -1 \\ a_1 &= 3 - 2^{1+2} = -5 \\ a_2 &= 3 - 2^{2+2} = -13 \end{aligned}$$

B. $a_n = 2na_{n-1}, a_0 = 3$

$$a_1 = 2 \cdot 1 a_0$$

$$a_2 = 2 \cdot 2 a_1 = (2 \cdot 2)(2 \cdot 1) a_0$$

$$a_3 = 2 \cdot 3 a_2 = (2 \cdot 3)(2 \cdot 2)(2 \cdot 1) a_0$$

$\vdots$

$$a_n = (2 \cdot n)(2 \cdot (n-1)) \dots (2 \cdot 1) a_0$$

$$= (2 \cdot 2 \dots 2)(n \cdot (n-1) \dots 1) a_0$$

$$= 2^n n! a_0$$

$$a_0 = 3(2^0)(0!) = 3(1)(1) = 3$$

$$a_1 = 3(2^1)(1!) = 3(2)(1) = 6$$

$$a_2 = 3(2^2)(2!) = 3(4)(2) = 24$$

Now plug in $a_0 = 3$:

$$\boxed{a_n = 3(2^n)(n!)} \leftarrow \text{Final closed form}$$

$$a_1 = 2(1)(3) = 6$$

$$a_2 = 2(2)(6) = 24$$